

Lesson 1: The Concept of Instantaneous Rate of Change

Topic 1.1: Introducing Calculus: Can Change Occur at an Instant?

As students of life, and algebra, you have studied many rates of change for different situations: how fast we drive, how our savings grow over time, the rate that the world population changes, our growth rate every year, unemployment rates, and so much more. In each of these studies, we can express one variable in terms of another, that is, as a function, $y = f(x)$.

Velocity

Consider an object that travels in a straight line. The **average velocity** over some given time interval can be defined as the ratio of the change in position (**displacement** or **net change**) to the elapsed time period:

Average Velocity =

However, when we think of velocity, we usually mean **instantaneous velocity**, which indicates the speed and direction of some object at a particular moment, which could be faster or slower than the average velocity.

Can you explain why we can't define instantaneous velocity as a ratio?

How can you estimate an instantaneous velocity?

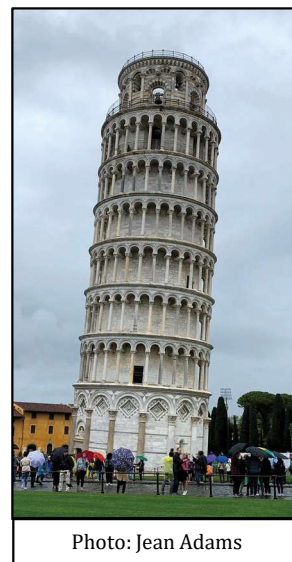
Let's generalize that big idea with this principle:

Average velocity over a very small time interval is very close to instantaneous velocity.

EX #1: If you climb to the top of the Tower of Pisa and drop a coin to the ground, estimate the instantaneous velocity at $t = 0.7$ seconds. Use Galileo's formula $s(t) = 16t^2$ to compute the average velocity, in feet/second, over the time intervals listed in the table below.

Time Interval	Average Velocity
[0.7, 0.71]	22.56
[0.7, 0.705]	
[0.7, 0.7001]	
[0.7, 0.70005]	
[0.7, 0.700001]	

What is your estimate for the ***Instantaneous Rate of Change*** (IROC) at $t = 0.7$ second?



THE STUDY OF RATE OF CHANGE IS FUNDAMENTAL TO THE STUDY OF CALCULUS.

In our first example, we allowed the time interval to shrink to zero, so can say that *the average velocity converges to the instantaneous velocity* or that the *instantaneous velocity is the limit of the average velocity*.

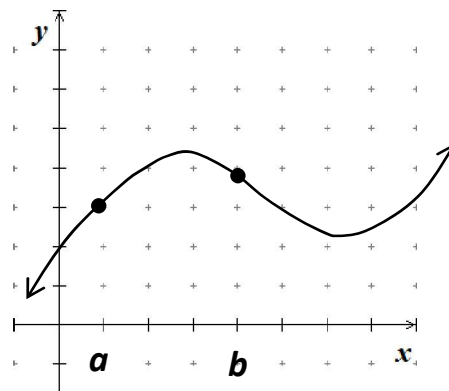
Graphical Representation of Velocity

Average Speed (AROC) is defined by the total distance traveled, d , divided by the elapsed time period, t .

Graphically, the average speed is the slope of the secant line.

Analytically, average speed $\text{AROC} = \frac{\Delta d}{\Delta t}$

Sketch the secant line through a and b . Estimate the average velocity.

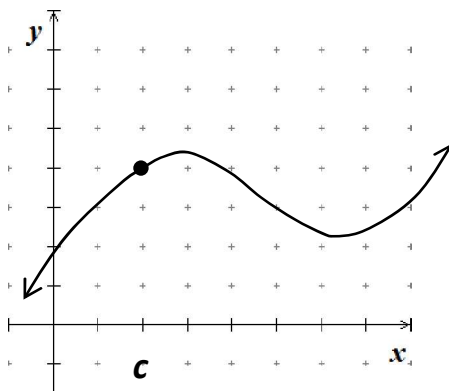


Instantaneous Speed (IROC) is the speed at a specific instant in time, where $t = t_0$ is the limit of the average rates of change.

Graphically, the instantaneous speed is the slope of the tangent line at a specific point.

Analytically, we need *calculus* to determine this instantaneous speed.

Sketch the tangent line through point c . Can you estimate the instantaneous velocity at $x = c$?



EX #2: Spin Class Heart Rate and Exercise

Have you ever taken a Spin Class at the gym? Then you know that after the class is over, your rapid heart rate decreases as time passes. Let $H(t)$ represent your heart rate. Write an expression for the rate of change of your heart rate over the period from $t = 0$ seconds to $t = 90$ seconds, after you have stopped exercising.



EX #3: Roller Coaster

A **hypercoaster** is a special circuit roller coaster with a height measuring greater than 200 feet. If a new hypercoaster is under design to have its largest drop on the circuit modeled by the equation below, where $d(t)$ is measured in feet, and time t , is measured in seconds.

$$d(t) = 1.87t^3 - 20.64t^2 + 112.79t + 132.56$$

- A. Find the average speed (velocity) of the hypercoaster from 2 seconds to 8 seconds.
- B. What is the average speed (velocity) of the hypercoaster from 4 seconds to 6 seconds?
- C. Estimate the instantaneous speed (velocity) of the hypercoaster at exactly 5 seconds.

Time Interval	Average Velocity
[5, 5.1]	
[5, 5.01]	
[5, 5.001]	

- D. Explain the meaning of the instantaneous speed (velocity) in the context of the question at $t = 5$ seconds.

EX #4: Airport Departures



The traffic pattern of departing flights at Orlando International Airport on a fall afternoon can be modeled by the function V defined by $V(t) = 92 - 15 \sin\left(\frac{t}{3}\right)$ where $V(t)$ is measured in vehicles and t is measured in minutes $0 \leq t \leq 30$.

A. Find $V(t)$ for $t = 18$.

B. Make a sketch of the graph of time vs. number of vehicles. Be sure your graph has a scale.



C. What is the average rate of change of the vehicles at the departing flights over the time interval $15 \leq t \leq 20$? Indicate units of measure.

D. Estimate the instantaneous rate of change of vehicles at $t = 18$ by finding the average rates from $t = 18$ to $t = 18.1$, $t = 18$ to $t = 18.01$, and $t = 18$ to $t = 18.001$.

Time Interval	Average Velocity
$[18, 18.1]$	
$[18, 18.01]$	
$[18, 18.001]$	

E. Why can't the instantaneous rate of change of traffic in the departure lane with respect to time be calculated using the method in part C?